
Information-Guided Structured Exploration

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Abstract

Organisms are intrinsically motivated to seek information about their environment, yet how this curiosity guides exploration remains poorly understood in both cognitive science and reinforcement learning (RL). Most curiosity-driven RL operationalize intrinsic motivation as novelty seeking, which leads to the noisy-TV failure. We argue that epistemic curiosity, which drives exploration towards theory-building rather than novelty would lead to different exploration behavior. To test this, we introduce a partially observable grid world where colored tiles emit stochastic signal that follow a probability distribution rule and a latent goal tile that must be discovered through exploration. We compare a novelty-driven and an epistemic-driven agents with different intrinsic reward function: prediction error versus expected information gain. We hypothesize that novelty-driven agents will spend more time around surprisal tiles and show less rule-consistent visitation patterns, whereas epistemic-driven agents will prioritize informative cues and more reliably infer the hidden sequence, improving performance across resets.

1 Background and Related Work

Intrinsic motivation is widely used in reinforcement learning (RL) when extrinsic rewards are sparse, delayed, or absent. In this setting, curiosity is typically modeled as an internal reward that encourages agents to gather information and build reusable knowledge about the environment [1].

A central divide is between novelty-driven and epistemic curiosity. Prediction-error methods, such as the Intrinsic Curiosity Module (ICM), reward agents for encountering hard-to-predict outcomes and have been shown to improve exploration in sparse-reward tasks [2]. However, such methods can be misled by stochastic but uninformative signals, as seen in the noisy-TV problem [3].

Information-theoretic approaches instead define curiosity in terms of belief updating. Still and Precup propose intrinsic objectives based on information gain, such as KL divergence, so that exploration is directed toward reducing uncertainty about latent environmental structure rather than merely chasing unpredictability [4]. This distinction motivates comparing prediction error and expected information gain in environments that separate informative cues from stochastic traps.

Recent work also emphasizes that curiosity should be evaluated not only by task performance but by the structure of exploration itself. Gruaz et al. show that the effectiveness of intrinsic rewards depends strongly on environment design, highlighting the need for controlled testbeds that isolate specific failure modes [1]. Similarly, InfoRL demonstrates that information-theoretic objectives can reveal interpretable differences in behavioral strategy, not just improve reward outcomes [5].

2 Research Plan and Timeline

2.1 Environment design

We model the grid-world as a Partially Observable Markov Decision Process (POMDP) defined by $(S, A, T, \Omega, O, R, \gamma)$. The latent state $s \in S$ includes the agent’s position in an $n \times n$ grid and hidden variables such as tile types and a latent rule governing a final puzzle. The agent navigates the grid using actions A , with transitions given by T . Because tile types are not directly observable, the agent receives observations $o \in \Omega$ generated by $O(o|s)$. Clue tiles emit informative signals about the hidden rule, whereas surprisal tiles produce random, uninformative signals. A special puzzle tile cannot be accessed through normal movement and instead requires executing the correct action sequence inferred from gathered clues. The agent must therefore maintain beliefs over hidden states and explore the environment to discover the rule and obtain reward R by solving the puzzle.

2.2 Agent

Based on [1], we here consider several reinforcement learning agents that differ in the intrinsic reward signal guiding their exploration behaviors. The novelty-driven agent uses prediction error as their reward (i.e. higher prediction error leads to higher rewards), whereas the information-theoretic driven agent uses expected information gain calculated through KL divergence (i.e. $\text{KL}(\text{posterior} \parallel \text{prior})$).

2.3 Training Algorithms

We aim to train the RNN model using an advantage actor–critic (A2C) or proximal policy optimization (PPO) algorithm. Based on [6], we add an additional information-theoretic constraints to the model as the intrinsic reward to the agent, which calculates the KL-divergence between the agent’s prior and posterior latent beliefs. This will allow us to test whether the main behavioral patterns of curiosity-driven exploration and it will help us separate effects that come from the reward design itself from those that depend on the complexity of the function approximator.

2.4 Behavioral Metrics

We will define and analyze evaluation metrics for structured exploration, such as success rate, time-to-solution, revisit frequency around trap tiles, and trajectory-level measures of behavioral organization. We hope that this extension will help us to distinguish genuinely informative cues from stochastic trap states such as “noisy-TV” tiles, which may generate high prediction error without improving the agent’s understanding of the environment.

Taken together, these components will allow us to establish the environment, a primary deep RL agent, an interpretable tabular baseline, and a more principled way to measure information-seeking behavior.

3 Author Contribution Plan

Yi, Jialin, and Mingze contributed equally to the conceptual development, environment design, implementation, experimentation, and writing of this project.

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